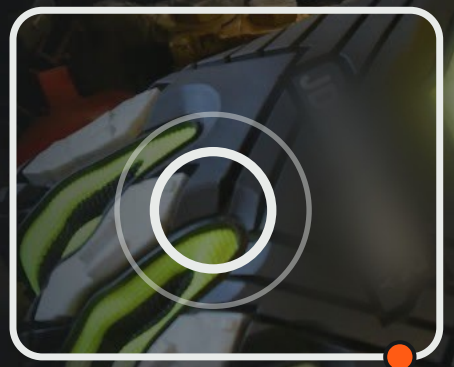
praxis

STEP 4/9

Instructions that can finally see how you're doing

An AI coach for new maintenance technicians. It watches the work through a first-person camera and says the next step.



Seat the clamp past the hose bead

CONCEPT MOCK · STOCK PHOTO

STARTUPS @ HARVARD · 20 OCT 2026 · MONISH SARAVANA

THE PROBLEM

A new maintenance technician needs about a year of a senior tech's time to work alone

9–14

months before a new technician pays for themselves, in a survey of 35 repair stations.

$10–15k

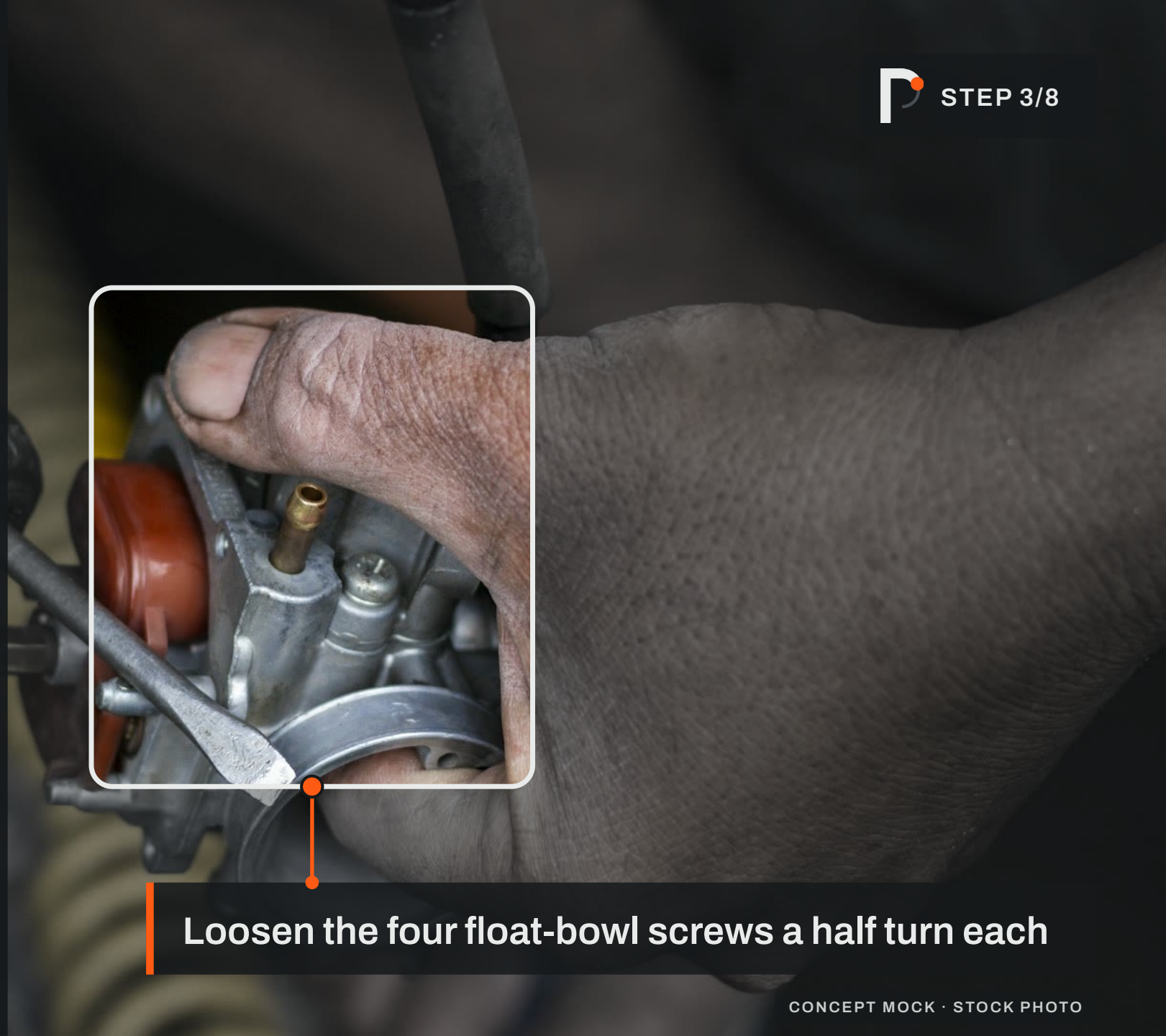
to hire and ramp up one maintenance technician, counting onboarding and lost productivity.

~255,000 openings a year for US maintenance and repair techs, and every new hire pulls a senior tech off the floor.

THE PRODUCT

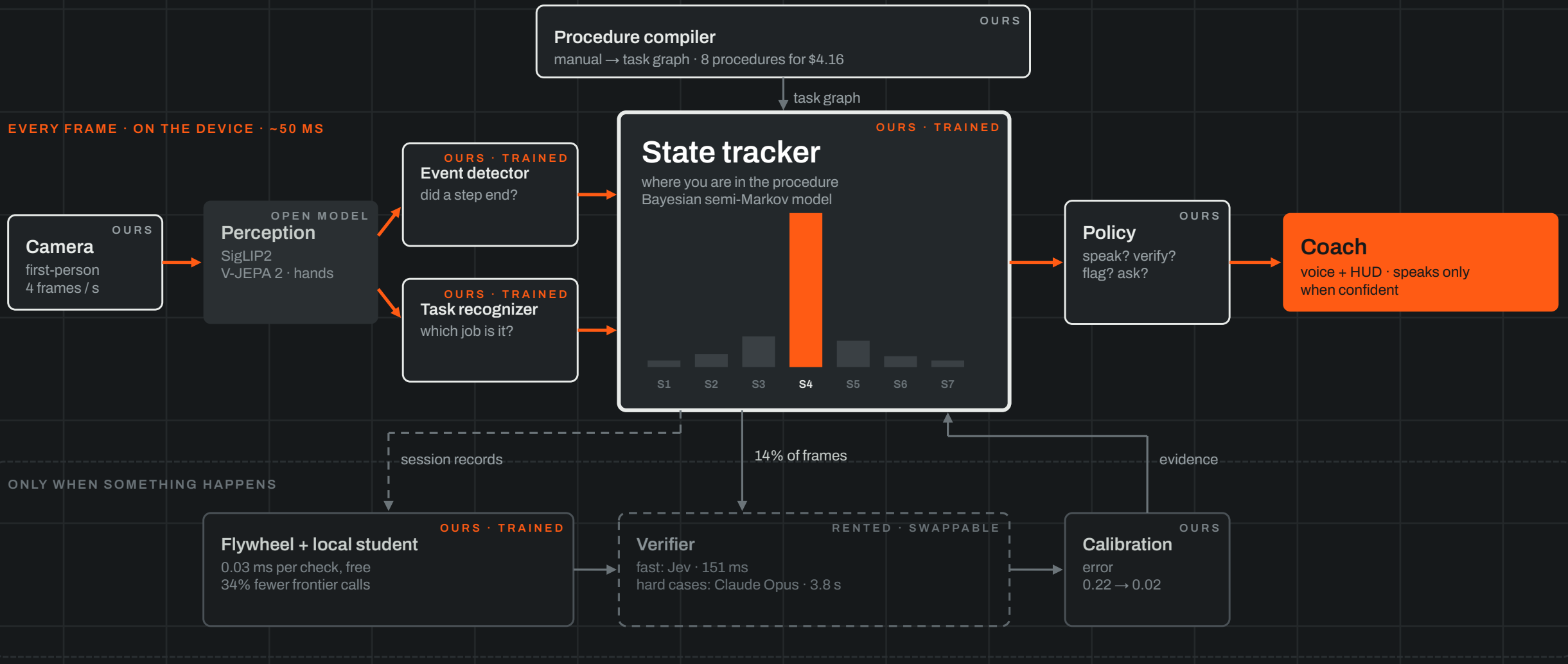
Praxis watches the work and coaches one step at a time

LOAD	the manuals and checklists a shop already has
COACH	through smart glasses, one short line at a time
VERIFY	critical steps and keep the record



HOW IT WORKS · ENGINE v0

The frontier model is one swappable part. The tracking engine around it is ours.



A shop uploads the manuals it already has. Praxis turns each procedure into a coach

3 DOCUMENTS

DOCUMENT	PAGES	SECTIONS	LIKELY PROCEDURES
FIST 1-1 · Hazardous Energy Control Program FIST_1-1.pdf · v1	97	131	4
FIST 4-1A · Maintenance Scheduling for Mechanical Equipment FIST_4-1A_2020.pdf · v1	222	234	3
FIST 3-8 · Protective Relays: Operation, Maintenance and Field Tests FIST_3-8_2021.pdf · v1	125	97	0

1. **Upload the manuals**
   A 97-page safety manual is indexed in 0.34 s on a laptop, for $0. Its procedures are found automatically.

2. **Review the coached steps**
   Each procedure compiles for about $0.19, with cards quoted from the manual. A manager approves it first.

3. **Coach through smart glasses**
   Live tracking and voice control, with Meta and Viture glasses connections and screen or voice-only modes.

High Voltage Board Repair (Electric Screwdriver)

v1 · approved · live · Approved · live

Source **High-voltage board repair (electric screwdriver)** v1 · High Voltage Board Repair Procedure (Electric Screwdriver) · Steps from split call · Sent 6,569 chars of a 1-page document
Local tracking: document only · mark steps on a recording · Cost $0.26 · claude-opus-5-5 · Oct 18

WHOLE PROCEDURE

- **high voltage** HIGH VOLTAGE BOARD REPAIR PROCEDURE (ELECT... ×
- **high voltage** Observe the high voltage board for a moment ×
- **high voltage** Take the high voltage board and places it on the work area ×
- **electrical: 5 Volts** Adjust the power supply's voltage knob to set it to 5 Volts ×
- **disconnect the power** Disconnect the power supply cables ×
- **temperature: 200 degrees** Set the soldering iron temperature to 200 degrees using th... ×
- **temperature: 160 degrees** Reduce the soldering iron temperature to the minimum (1... ×
- **temperature: 160 degrees** Lower the soldering iron temperature to the minimum (160... ×
- **electrical: 5 Volts** Set the voltage knob of the power supply to 5 Volts ×

STEPS 20 · 0 safety-critical — Expand all

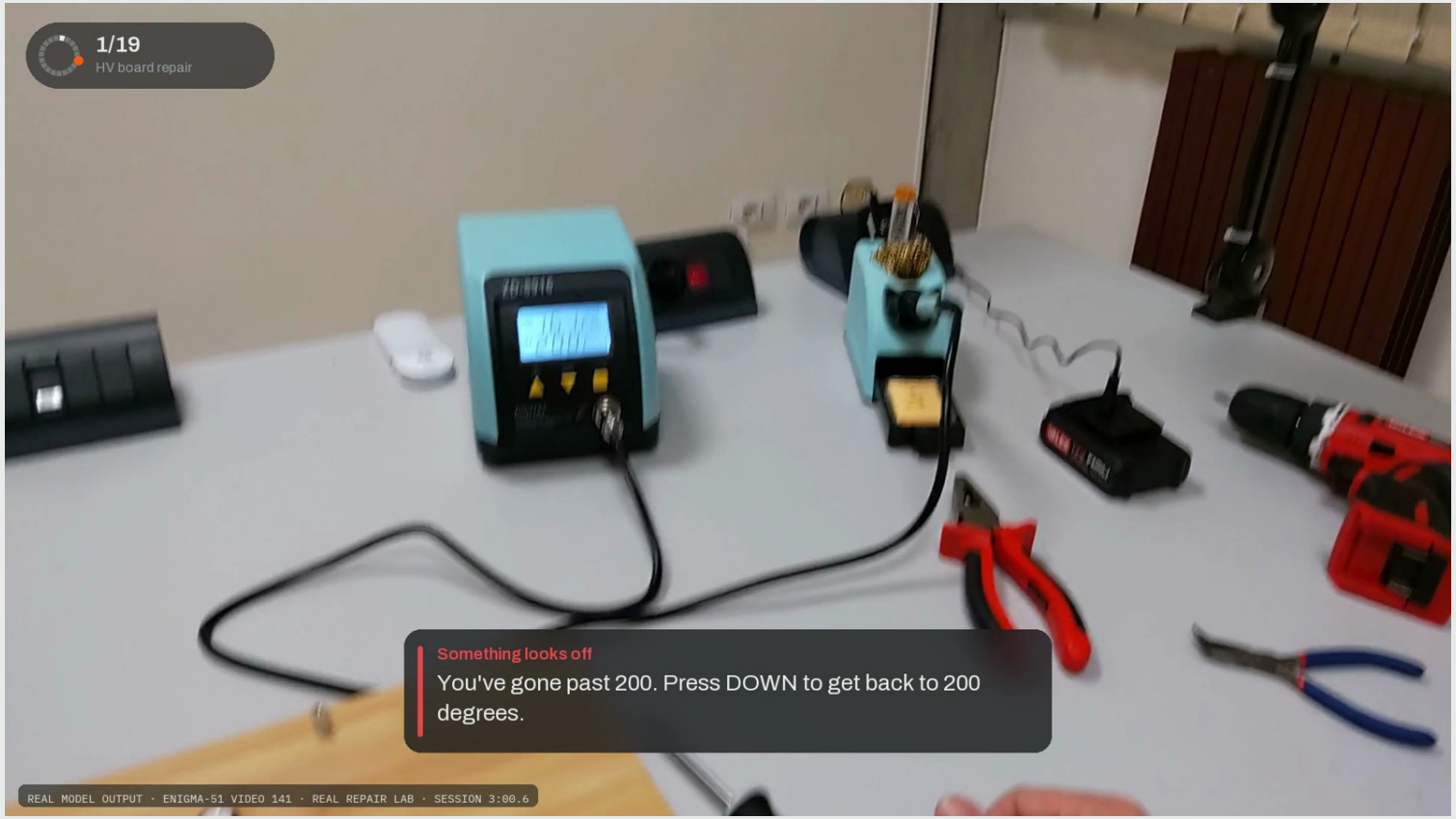
1. **Start recording and toggle bench lamp** — Sit at the workbench and say "Record" clearly, then wait for the b... — Safety wording in source — 1 card
2. **Press green buttons on Panel A** — Walk over to Panel A and look at it for a moment. Then press both green buttons.
3. **Place board and connect screwdriver battery** — Go back and sit at the bench, look at the high voltage board, th... — Safety wording in source — 2 cards
4. **Secure board to workbench with screwdriver**

What the compiler saw — 6,569 chars

Page 1 ↗

HIGH VOLTAGE BOARD REPAIR PROCEDURE (ELECTRIC SCREWDRIVER) Sit at the workbench Pronounc the voice command "Record" to start the recording; you will hear an audible signal for confirmation Turn on and off the lamp on the workbench while looking at it Leave the laboratory Enter the laboratory and closes the door Head towards Panel A and observes it for a moment Press the two green buttons on Panel A Move back to the workbench and sits down Observe the high voltage board for a moment Take the high voltage board and places it on the work area Connect the battery to the electric screwdriver Remove the screws from the workbench using the electric screwdriver Secure the board to the workbench using the electric screwdriver Observe the power supply Turn on the power supply using the socket switch Adjust the power supply's current knob until the green LED lights up Adjust the power supply's voltage knob to set it to 5 Volts Connect the power supply cables to the board's power points Watch the board for a few seconds to check if the red and green LEDs turn on Turn off the board's power supply using the socket switch and wait for the board's LEDs to turn off Set the current and voltage knobs of the power supply back to 0 Disconnect the power supply cables Remove the board from the workbench using the electric screwdriver Observe the soldering iron Turn on the soldering iron using the socket switch Set the soldering iron temperature to 200 degrees using the yellow "UP" button Grab the capacitor on the board with pliers Take the soldering iron's probe Touch the first pin of the capacitor with the soldering iron's probe for 5 seconds Touch the second pin of the capacitor with the soldering iron's probe for 5 seconds Place the pliers on the workbench Place the soldering iron's probe Position the board vertically Grab

A live session on a real repair job. Every card on screen is the model's own output



0:00	Starts from the lab's written procedure. It was never trained on this task.
0:26	The fast model suspects a missed button. Opus checks and clears it, so no false alarm.
2:52	Safety-critical step: it reads the iron at 215 °C and tells the tech to go back to 200.
3:37	Follows the resolder, step by step.

ENIGMA-51 · REAL REPAIR LAB · 16-MINUTE LIVE SESSION · $0.17

HOW FAR ALONG · MEASURED

It already beats a frontier model on its own. The bottleneck is maintenance data

13% more

accurate at tracking the step than the same frontier model called every 5 seconds

1/5 the cost

$0.018 vs $0.085 per minute of video, because it only asks a model when something happens

11 of 15

technician questions on an industrial repair job answered correctly from the company's own manuals

0

times confidently wrong when the view was blurred or blocked. It asks for a better view instead

THE BOTTLENECK: DATA

First-person video of real maintenance work isn't sold off the shelf. So we collect it:

- Pay students at trade schools to wear glasses on real jobs
- Label it with our own step-marking tool
- Every pilot adds more, with the customer's consent

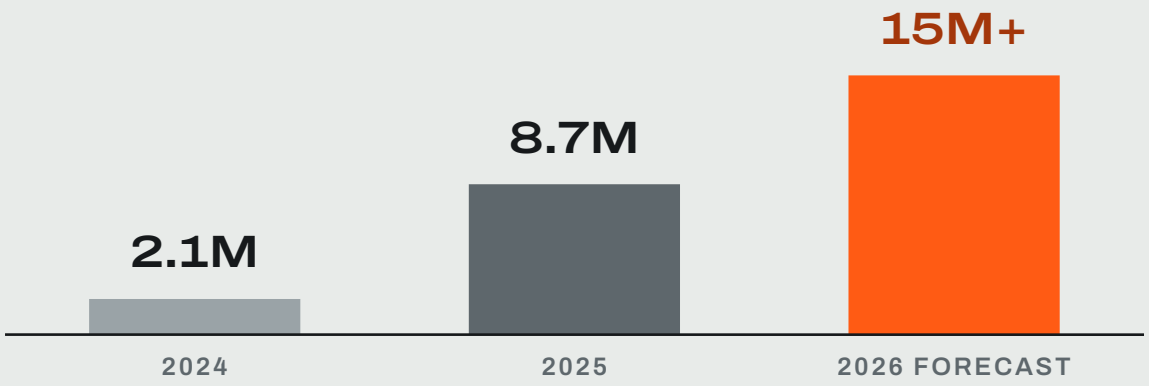
About 80 hours of footage for $2,000, then GPU time to retrain.

WHY NOW

Smart glasses took off, and AI keeps getting cheaper

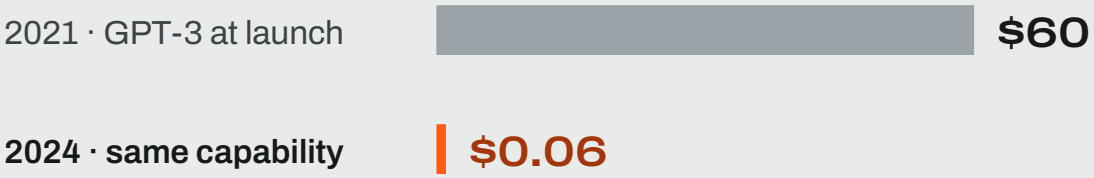
AI GLASSES SHIPPED A YEAR

7× more glasses in two years. Every pair is a camera we can run on



PRICE OF THE SAME AI, PER MILLION TOKENS

1,000× cheaper in three years, about 10× every year



So a coach that watches a whole shift is finally affordable.

MAY 2026
Autodesk buys MaintainX, a maintenance app, for $3.6B

SEPT 2026
Meta opens its glasses' camera to outside apps

NOV 2026
Android XR glasses reported to ship

Q1 2027
Viture + NVIDIA coaching glasses ship

COMPETITORS

Rolling out a new procedure: what it takes today, and what it takes with Praxis

Procedure apps MaintainX, Augmentir	Type the steps into the app	The tech reads them on a phone	Ticks a box when done	Nobody sees whether the step was done right
Fixed cameras Retrocausal, Invisible AI	Mount a camera at one workstation	Record and label runs to train it	It watches that one station	Can't follow a tech around a machine
First-person AI glasses Strivr, Viture + NVIDIA	Their team films your procedures	They train a model for your site	Coach through their glasses	Filming and training for every customer
Praxis	Upload the manual you already have	Approve the coached steps, about $0.19 each	Coach through smart glasses	**Verified and recorded, with no filming**

WHY NOBODY OWNS THIS NICHE

Meta, Google and Apple won't specialize, so we will. Their glasses become our hardware

15M+

AI glasses forecast to ship this year. Every pair is a camera we can run on, and we pay $0 to build it.

Sept 2026

Meta opened its glasses' camera to outside apps. The platforms want specialists building on top.

$0

What Meta AI and Gemini Live charge consumers to look through the camera and help. We don't sell against free.

THEY BUILD FOR EVERYONE

Glasses, cameras and one free general assistant that answers questions about a single frame.

WE BUILD FOR ONE TRADE

A coach for maintenance teams: their own procedures, step-by-step state, verified records, a pilot on site.

That's also why we don't start with consumers. Consumer help is their turf and they give it away free. In a maintenance shop someone pays, and the same checklist runs every month.

MARKET + PRICING · US, PER YEAR

A $306M wedge in a $1.8B first market, in a $9B one

TAM — $9.2B

15.4M US workers who follow written procedures

SAM — $1.8B

2.47M US maintenance and repair techs

WEDGE — $306M

~255k new maintenance techs a year, coached

YEAR 3 — $4.5M ARR

~3,750 new techs, about 150 teams

HOW WE CHARGE

A short paid pilot
with a metric agreed up front. The fee counts toward year one

$200 per new-tech seat a month
billed annually, unlimited sessions. When they work alone, the seat moves to the next hire

One-time setup per procedure
we compile and check the shop's own manuals

$50 a month for every tech, next
skills records for the whole team

SAVINGS + MARKETING PLAN

A new tech costs $10–15k to hire and ramp up. A seat is $200 a month. One saved hour covers it

$10–15k

to hire and ramp up one maintenance tech today

$200

a month per seat for each new tech, billed annually. Unlimited sessions; the seat moves to the next new hire

1 hour

a week of a senior tech's time ($44) is what a seat costs, so saving one hour a week pays for it

MARKETING + SALES PLAN

OUTREACH → DEMO → PAID PILOT → ANNUAL SEATS → EVERY SITE

1 Direct outreach

I email and call maintenance managers at plants, facilities teams and repair shops. 70 this week.

2 Trade schools

HVAC and maintenance programs coach students with Praxis. Graduates bring it to their first job.

3 Trade shows + forums

Maintenance and reliability conferences, and the online forums where techs and managers trade advice.

4 Partners

Built into the maintenance apps shops already use, and resold by distributors and glasses makers.

THE LONG GAME

Every coaching session builds data nobody else has. That data is how we climb from technicians to robots

NOW · 2027

Coach new techs

Paid pilots with maintenance teams. Every session becomes labeled training data.

$0.3M · 10 teams

NEXT · 2028

Our own models

Trained on our pilot footage, on the glasses. Cheaper to run and better at maintenance than rented models.

$1.5M · 50 teams · 5×

THEN · 2029

A skills record for every tech

Verified work and certifications live in Praxis, so leaving means losing them. Then new trades.

$4.5M ARR · 150 teams · 3×

LATER · 2030+

Robots: two paths

A Train our own robot models on the data we've collected

B Sell the data and our step checker to robot labs

$13M → $30M ARR, 2030–31

AI LABS ALREADY PAY FOR DATA, AND IT'S GROWING

$2B+ a year

Mercor's run-rate selling expert data to AI labs, doubled in four months

>$100M a year

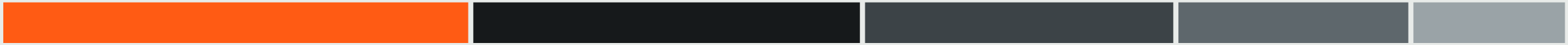
what robotics companies already spend buying real-world data

$8.6B by 2030

the AI training-data market, growing about 22% a year

THE ASK

$10k buys the data and GPU time to make the models better, and puts Praxis on glasses at a first pilot



$3,000

GPU time

Train our models (step tracker, event detector, local checker) on more data

$2,500

Glasses + capture

Run Praxis on camera glasses at a pilot site

$2,000

Training data

Buy labeled first-person recordings of real maintenance work

$1,500

Inference

About 1,400 hours of live pilot coaching, at ~$1.08 an hour

$1,000

Company formation

So a customer can sign a paid pilot

TRUE BY APRIL

Models retrained on real maintenance recordings

Running on glasses at a real site

A first paid pilot, with a metric agreed up front

INFO SLIDES

Appendix

A1 Evidence: accuracy and cost

A2 Under the hood: the live session

A3 What we train, and why big labs won't

A4 The status quo, tool by tool

A5 Market math and assumptions

A6 Unit economics and pricing risk

A7 Learning curves

A8 Training data, stated plainly

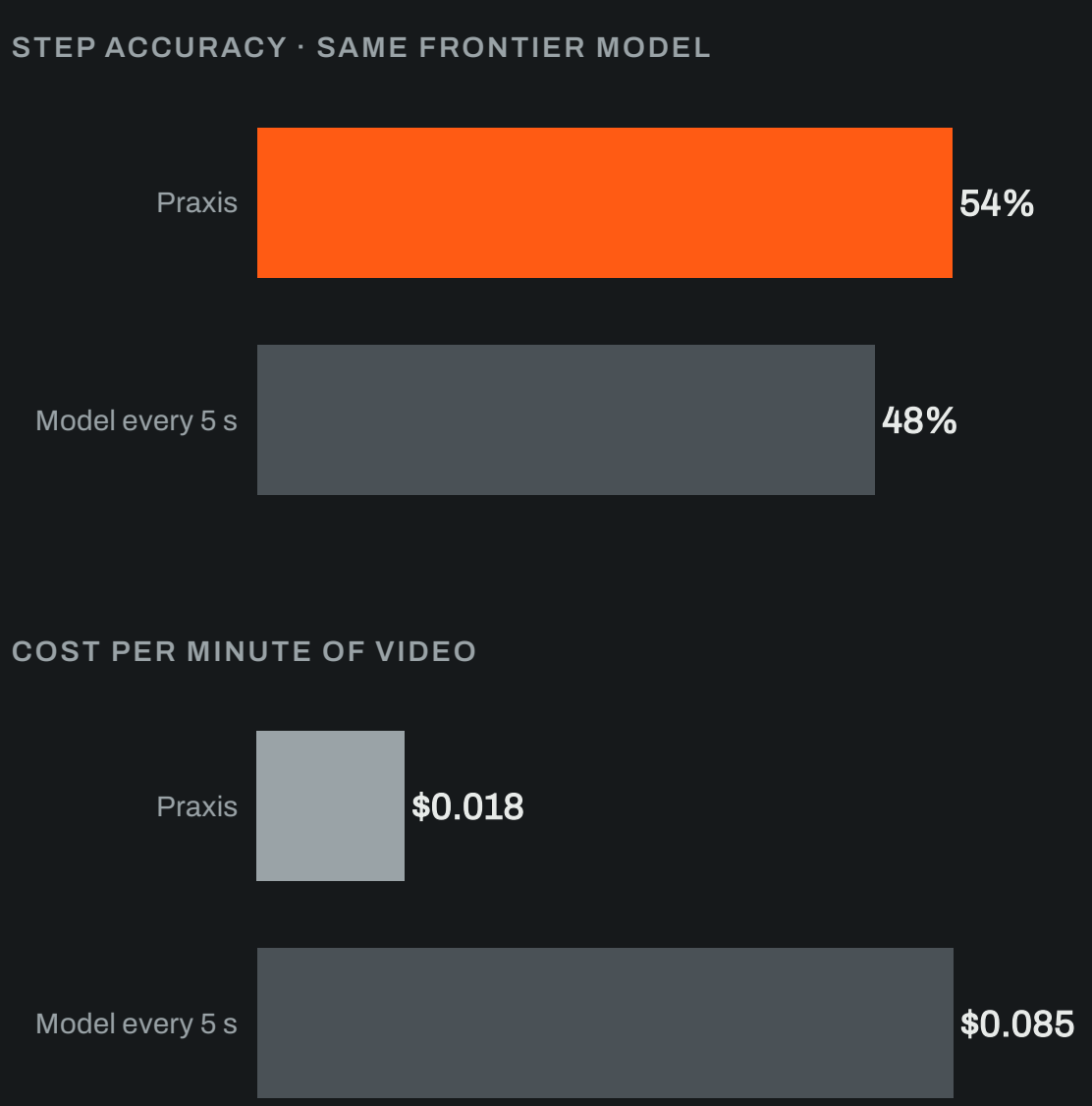
A9 Competitor detail

A10 Why not consumers first

A11 Risks and what we do

A12 Glasses and voice

A13 Why me



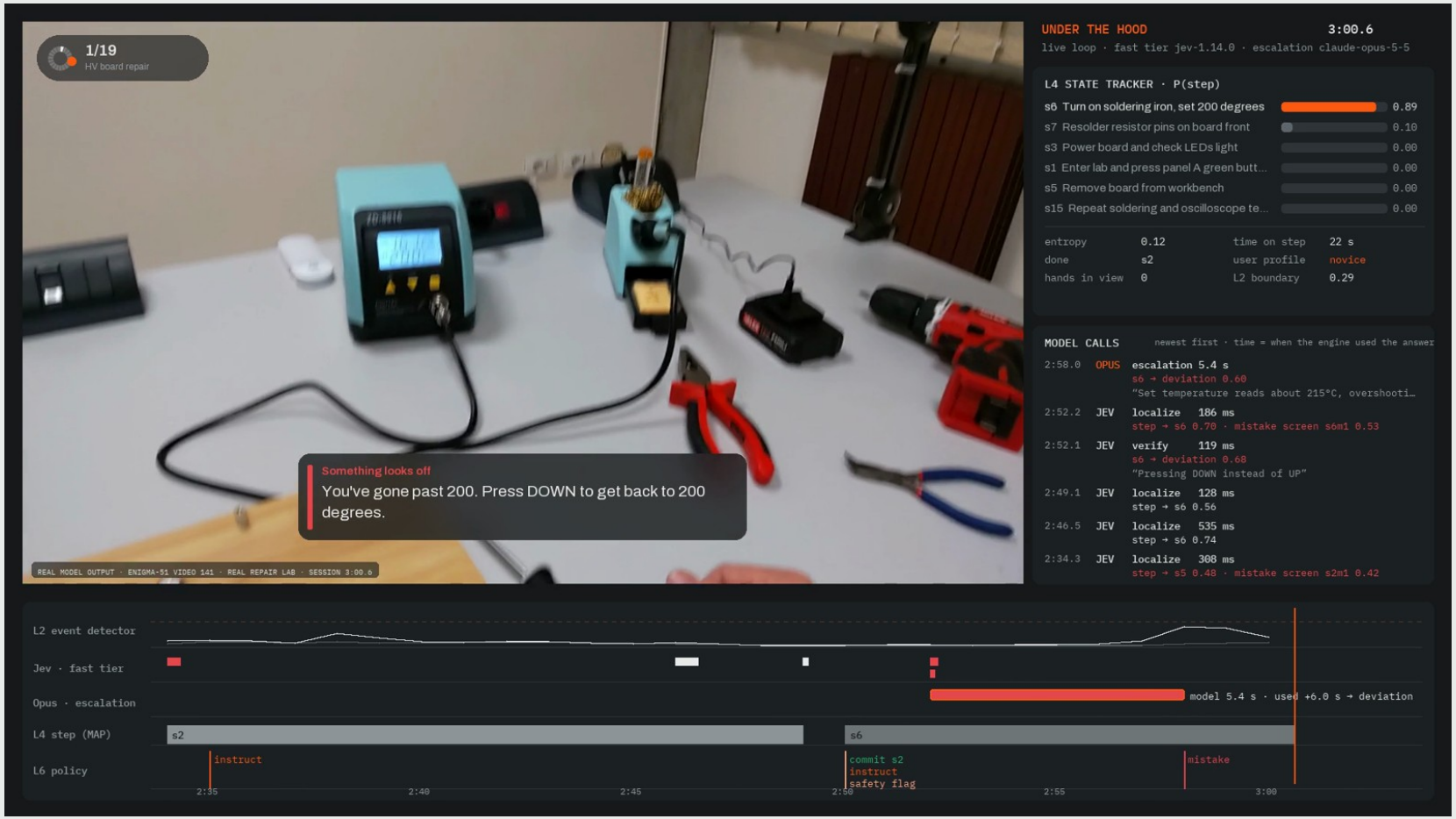
APPENDIX · A1

Same frontier model, with our tracking: 13% more accurate at a fifth of the cost

1/5

the cost per minute of video, and more accurate.

Under the hood: the same live session, layer by layer



State tracker. Its belief over the 19 steps, the time on the current step, and whether the tech looks like a novice.

Model calls. Jev answers in 150 ms (median, 130 calls). Opus double-checks 14 moments, about 4 s each.

Timeline. The event detector, both model tiers, the tracked step and every decision to speak or stay quiet.

Cost. $0.165 for the whole 16:40 session, with 0 errors.

APPENDIX · A3

What we train in-house, and why a big AI lab won't build this

PART	WHO	WHAT IT DOES
Event detector	OURS · TRAINED	A small temporal network that spots the moment a step ends
Task recognizer	OURS · TRAINED	Works out which procedure the tech is doing
State tracker	OURS · TRAINED	A Bayesian model of where the tech is in the procedure, with step models per procedure
Local checker	OURS · TRAINED	Learned from the frontier model; answers routine "is this step done?" checks for free
Perception	OPEN MODELS	SigLIP2, V-JEPA 2 and hand tracking turn each frame into features. Not trained by us
Verifier	RENTED	A frontier model answers hard questions on 14% of frames. Swappable in one line

WHY A BIG LAB WON'T

They sell one general model. Ours wraps any model, so their best one becomes our swappable verifier.

The hard part is data they don't have: first-person video of skilled work, lined up to the procedure, with verified steps.

And a business they don't run: each shop's manuals and records, and a pilot on site.

The status quo: what each tool does today, and what it misses

SHADOWING

A senior tech, side by side

The real teacher today

MISSES About $44 an hour of senior time, one new tech at a time

MANUALS + VIDEOS

PDFs, binders, YouTube

Explain what should happen

MISSES Can't see what you're actually doing

PROCEDURE APPS

MaintainX, Tulip, Augmentir

Put the steps on a phone; collect sign-offs and photos

MISSES A checkbox isn't proof the step was done right

FIXED CAMERAS

Retrocausal, Invisible AI

Watch one workstation on a line

MISSES Maintenance moves around the machine

PRAXIS

Watches the work, follows the person around the machine, and runs on hardware a shop already owns.

Market math, and the assumptions behind it

	WHO (BLS)	MATH	PER YEAR
TAM	Installation, maintenance + repair 6,086,190 · electricians 757,220 · production 8,565,760 = 15.4M	15.4M × $50 a month × 12	≈ $9.2B
SAM	General maintenance + repair, machinery mechanics + millwrights, HVAC = 2.47M jobs, ~255k openings a year	255k × $200 × 6 months = $306M; 2.47M × $50 × 12 = $1.48B	≈ $1.8B
SOM (year 3)	1.5% of new maintenance techs ≈ 3,750 new techs a year, ~150 mid-size teams	3,750 × $200 × 6 months	≈ $4.5M ARR

Prices are hypotheses. $200 a month for coaching anchors on Tulip's $100–250 per station; $50 a month for every-worker records is a guess

Openings include people changing jobs, so new-tech counts are an upper bound

Not counted: other countries, field service outside these groups, aviation maintenance, and the robotics upside

APPENDIX · A6

Unit economics, and the pricing risk

DESIGN	$ / VIDEO-MIN	$ / SEAT-MONTH, 2 H/DAY	MARGIN AT $200
Frontier model every 5 s	$0.085	about $214	negative
Praxis engine today	$0.018	about $45	about 77%
Praxis + our local model	$0.008	about $20	about 90%

Labeled assumptions, not sourced: if a senior tech spends 50% of their time on a new hire for 3 months, that's about $11,400 of their time; at 25% for 12 months, about $22,800. We're asking maintenance managers for their real split.

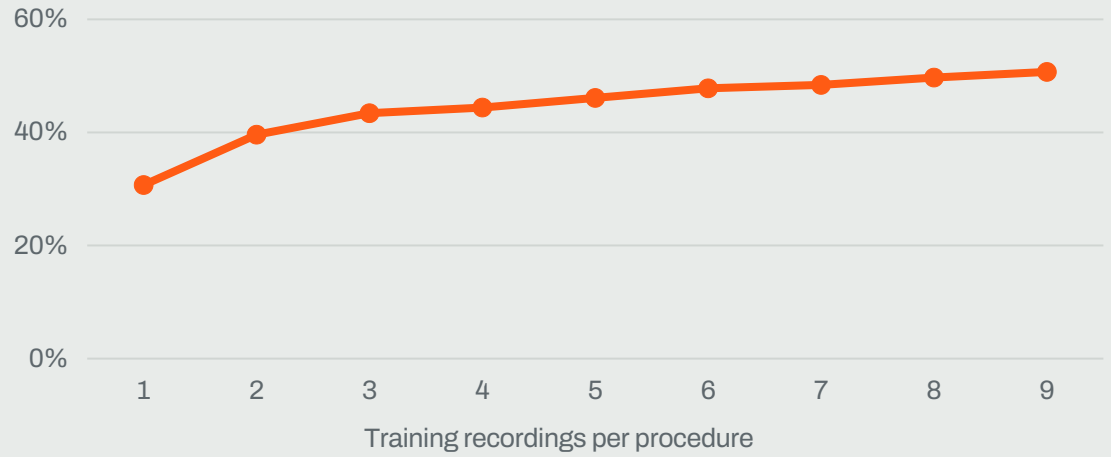
THE RISK

All-shift use (8 h/day) makes a flat $200 seat $1.19 an hour against our $1.08 cost: about a 9% margin. With the local model, about 60%. So price by coached hours or tiers, and move routine checks to the local model first.

Both learning curves: step models and the local student

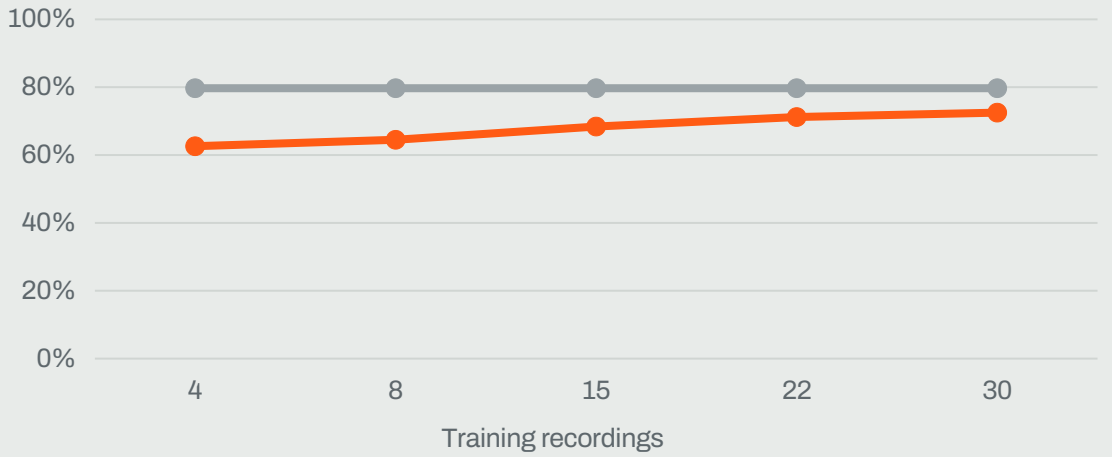
30.7% → 50.7%

recognizing the current step, on videos it never trained on



62.6% → 72.5%

checking a step is done · grey: the frontier model, 79.7%



TESTED ONLY ON RECORDINGS IT NEVER SAW · BOTH STILL RISING AT THE LARGEST SIZE WE HAVE

Training data, stated plainly

Train and test never overlap. Everything learned uses only the dataset's official train-split recordings

Small on purpose. A weekend-scale proof: the step-change detector trained on 24 recordings, the step models on 3 per procedure, the student on checks from 30

Tested on 48 held-out clips: 235 min, 302 annotated steps, 170 annotated mistakes, plus procedure types never seen in training

Design choices were made on train-split recordings only, never on the test clips

Labels are the dataset's, not ours. One run per system, no confidence intervals yet

LEARNING CURVES

Step models: 30.7% with 1 recording → 43.4% with 3 → 50.7% with 9 (the 3 procedures with 9)

Student: 62.6% with 4 recordings → 72.5% with 30; the frontier teacher is 79.7%

Detector: noisy, no clear trend yet. Not claimed

APPENDIX · A9

Who else is here, and how we differ

CATEGORY	EXAMPLES	US VS THEM
Instruction platforms	Tulip, Augmentir, Dozuki	They show the steps; we watch the work. Natural integration partners
Line cameras	Retrocausal, Invisible AI, Tulip Vision	Fixed at a station on a production line. Can't follow a person around a machine
First-person coaching	Viture Helix + NVIDIA, Strivr	Closest. Hardware-led or high-touch; no published accuracy
Toolkits	NVIDIA XR AI, Meta's camera toolkit, Android XR	Complements: the plumbing we build on
General assistants	Meta AI, Gemini Live	Answer questions about a frame. No plan, no state, no calibration

If a big platform ships a coach: they're building our hardware and plumbing. We go deep in one vertical, with its procedures, records and data.

Why not consumers first? Because the budget and the repeat data are in business

$3.6B

what Autodesk agreed to pay for MaintainX, an app that puts maintenance procedures on a phone.

$99 → $25

what Frontdoor's expert video help for homeowners fell to, per year, from 2023 to 2024.

The same checklist runs on every machine, every month. Home tasks don't repeat, and the platforms give consumer help away free. Consumers come last.

The risks we see, and what we do about each

RISK	WHAT WE DO
Meta or Google ships a coach	They're building our hardware and plumbing. We go deep in maintenance with its procedures, records and data, and run on their glasses
Glasses aren't ready	We run on glasses that ship today (Meta, Viture + NVIDIA). Each new device is one more adapter
Mistake detection is weak (4 of 17)	Locked procedures, maintenance data and the tech saying "done" all add evidence. Measured in every pilot
Privacy, unions, monitoring laws	Fast layers run locally; only event keyframes leave (14% of frames). Coaching for the worker, aggregates for the employer. Blur next
A wrong prompt on a safety-critical step	Lockout and similar steps stay the tech's job. Praxis prompts and records them, and never approves them
Depending on model vendors	The verifier is swappable in one config line, and our local student already handles routine checks
Solo founder	The grant tests demand before hiring. First hires come after the pilot

Glasses and voice: what's built, and what's next

BUILT

- Runs live in real time, with the local layers at about 50 ms per frame; model calls never block the loop
- Glasses connections for Meta and Viture + NVIDIA, plug and play
- Voice control, plus screen or voice-only modes depending on the glasses
- Praxis Studio: the engine in a browser, on a camera, an upload or a test clip

NEXT, WITH DEVICES IN HAND

- Test end to end on Meta and Viture glasses at a pilot site
- Measure glass-to-coach latency, the biggest unknown
- A hearing-protection headset for loud plants
- Face and badge blur; a local-model tier for sensitive sites

I've built real-time vision on messy video before, and I built this in a week

LipTrac

Real-time lip reading from ordinary video. 60,000+ frames, 12 testers, provisional patent filed.

7 years

of competitive robotics, coaching teammates without seeing what they see.

1 week

to build the engine, a 48-video eval and a live app, solo. First commit Oct 17.

Monish Saravana · Harvard College · solo founder. Also: path planning for oil-cleanup boats from aerial imagery.